## System-Level Patterns for Production

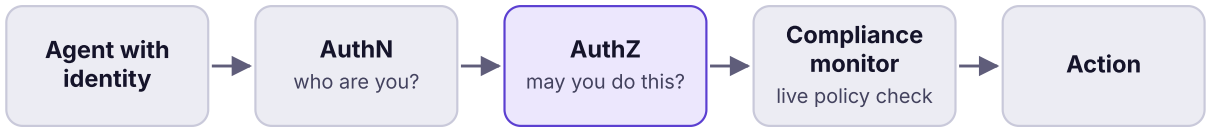
### 1. The plumbing around the agents

- **Tool and Agent Registry**: a discoverable catalogue of tools and agents (capabilities, schemas), so the system binds to the right one at run time instead of hard-coding (e.g. procurement discovering supplier agents). The enterprise version of the A2A agent card
- **Real-Time Compliance Monitoring**: every action checked against policy as it happens, blocked or flagged live (e.g. a healthcare agent screened for patient-data leakage)
- **Agent Authentication and Authorization**: agents have identities and scoped permissions; each request is authenticated and authorised before it runs (least privilege)
- **Event-Driven Reactivity**: agents wake on events (messages, webhooks, state changes), not only on request (e.g. a support agent woken when a ticket arrives)

### 2. Identity is the foundation of agent security

Once agents act autonomously and call tools, 'which agent is allowed to do what?' becomes a first-class question. Authentication + authorisation + real-time compliance monitoring let autonomous agents operate without exceeding their authority.

✎ DIAGRAM




Every agent action passes identity, permission and live policy checks.

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*"Agent authentication, scoped permissions and real-time compliance monitoring are what let agents act autonomously without exceeding their authority."*